

Head-restricted relaxed speculative decoding: final results

Bill Xu · 2026-09-17 · GPT-OSS-20B + EAGLE3 drafter · vLLM 0.26.0 · one H100 (Nibi) · 3 seeds per point

benchmark	problems x seeds	lossless accuracy	best tok_lt α	accepted per pass	answer length	verifier passes (speedup)	accuracy	best tok (α : speedup)
GSM8K	150 × 3	96%	0.15	1.12	1.02	1.07	96%	0.25: 1.11
AIME24	30 × 3	78%	0.15	1.13	1.01	1.08	79%	0.25: 1.05
HumanEval	150 × 3	96%	0.25	1.11	1.00	1.09	97%	0.25: 1.01
LiveCodeBench	90 × 3	89%	0.15	1.11	1.02	1.06	89%	0.15: 1.01
MT-Bench	80 × 3	—	0.25	1.11	0.99	1.10	—	0.25: 1.04
LongBench-v2	30 × 3	60%	0.2	1.11	0.99	1.09	53%	0.25: 1.09

Best tok_lt setting per benchmark (highest verifier-pass reduction among $\alpha = 0.15 / 0.20 / 0.25$). Every number is the rule ÷ lossless on the same problems and seeds; lossless = 1.00.

How to read this

- accepted per pass: drafted tokens the big model kept per checking pass (the gain the source papers report).
- answer length: mean tokens per answer ÷ lossless (the cost the source papers did not measure).
- verifier passes: lossless passes ÷ rule passes — the end-to-end speedup; one pass = one big-model step.
- budget hits: runs that ran to the token limit (looping / endless hedging).

What it shows

- On math, code and chat, tok_lt accepts 9–13% more per pass than lossless (tok: 2–6%), answers stay at lossless length, accuracy is unchanged, and the big model does 6–10% fewer passes. tok, the paper's rule, gains little end to end.
- LongBench-v2 (long-context QA) is the exception: no consistent speedup for either rule, and every relaxed setting is 3–8 accuracy points below lossless (within seed noise individually, but uniformly below).
- Rambling is caused by letting through tokens the verifier ranks far below its own favourite; tok and tok_lt never do, which is why their answers do not grow. Keeping that filter caps the gain at roughly 1.1× (see RESULTS.md).

Rules

- lossless: keep a drafted token with probability p/q (output identical to the big model alone).
- tok (Narasimhan et al. 2025, token variant): free pass for tokens within α of the verifier's top choice; stricter than lossless for all other tokens; residual from its blended distribution.
- tok_lt (ours): same free pass; ordinary lossless check for all other tokens; lossless residual.

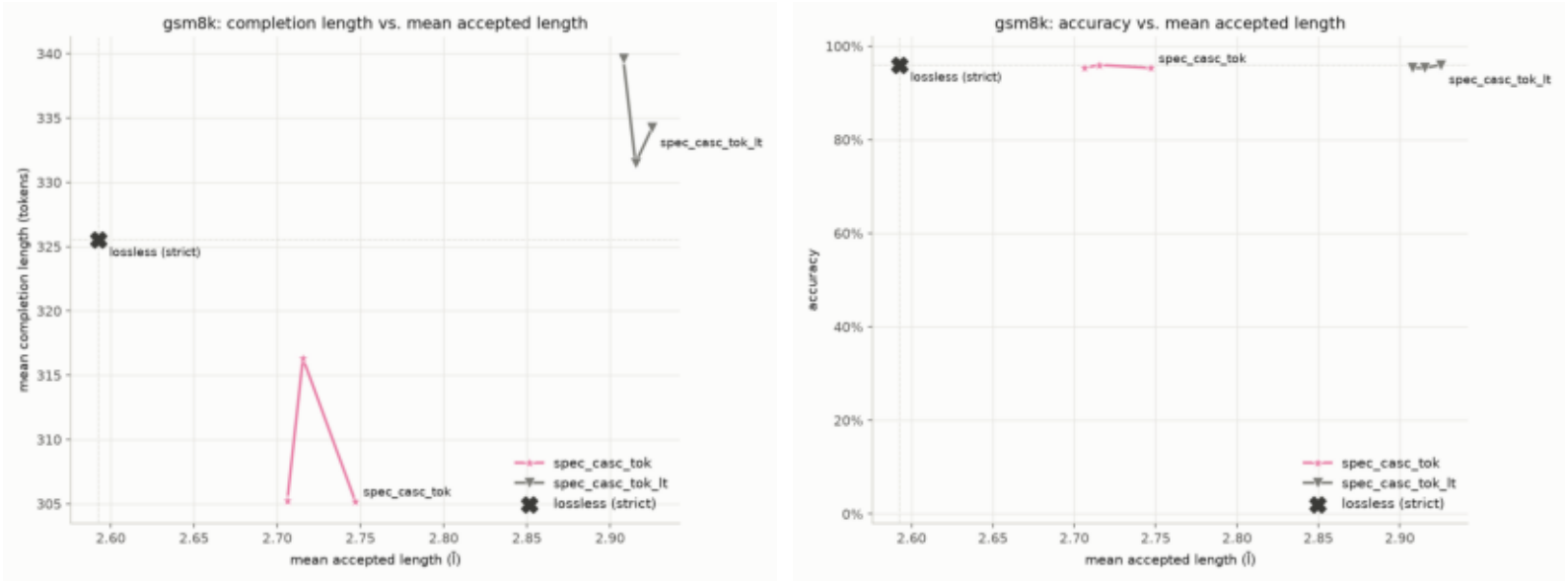
Protocol: warm vLLM server per (rule, α , seed); 3 seeds; problems per benchmark as listed; token budgets gsm8k 2,048, AIME24 32,768, HumanEval 9,000, LiveCodeBench 12,000, MT-Bench 4,096, LongBench-v2 8,192. Accuracies graded per seed. Detail and caveats: cascade/RESULTS.md, cascade/JOURNAL.md.

GSM8K — grade-school math

150 problems × 3 seeds = 450 runs per point · budget 2,048 tokens · relative to lossless, paired per problem and seed
lossless baseline: 2.59 accepted per pass, 326 tokens per answer, 7 of 450 runs hit the budget, accuracy 96%

rule	α	accepted per pass	answer length	budget hits	verifier passes (speedup)	accuracy
lossless	—	1.00	1.00	7 / 450	1.00	96%
tok (paper's rule)	0.15	1.05	0.97	7 / 450	1.06	96%
tok (paper's rule)	0.25	1.06	0.94	10 / 450	1.11	96%
tok_lt (ours)	0.15	1.12	1.02	8 / 450	1.07	96%
tok_lt (ours)	0.2	1.12	1.04	10 / 450	1.04	96%
tok_lt (ours)	0.25	1.13	1.03	11 / 450	1.07	96%

Shaded row: best tok_lt setting (fewest verifier passes). tok rows: the paper's rule at the tightest and loosest of the three settings.



Left: mean answer length vs mean accepted length per pass (x-axis), all settings of both rules; lossless is the x. Right: accuracy vs the same x-axis.

Short answers; the 7-11 budget-hit runs per point carry ~10% of all tokens, so per-seed speedups swing 0.93-1.22 even at 450 runs. Both rules tie at ~1.05-1.10x.

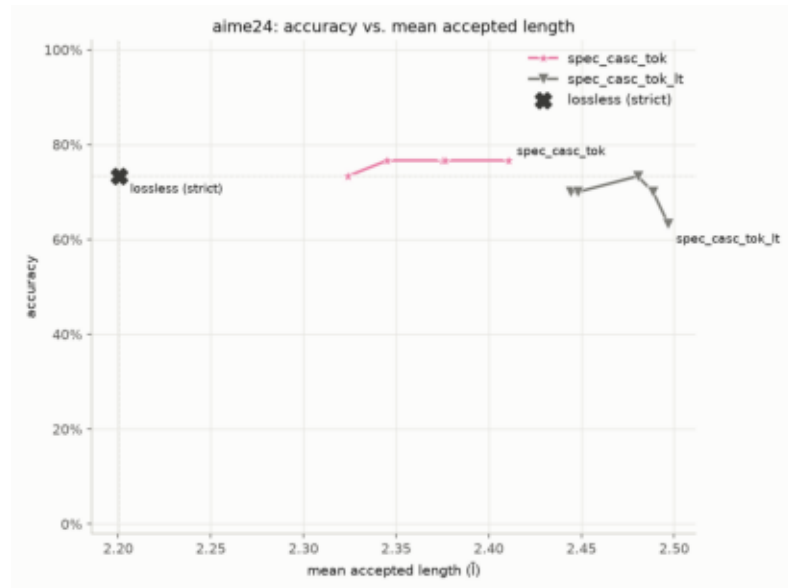
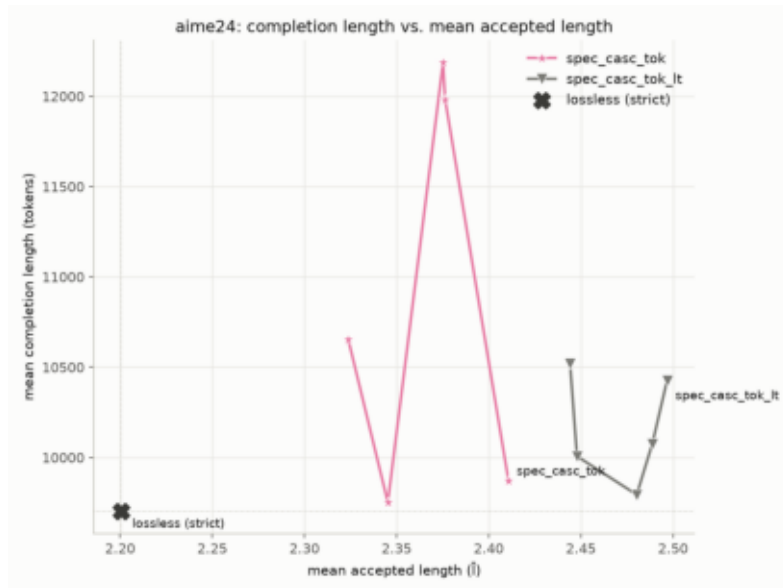
AIME24 — olympiad math, long reasoning

30 problems × 3 seeds = 90 runs per point · budget 32,768 tokens · relative to lossless, paired per problem and seed

lossless baseline: 2.20 accepted per pass, 9,703 tokens per answer, 6 of 90 runs hit the budget, accuracy 78%

rule	α	accepted per pass	answer length	budget hits	verifier passes (speedup)	accuracy
lossless	—	1.00	1.00	6 / 90	1.00	78%
tok (paper's rule)	0.15	1.08	1.26	11 / 90	0.84	78%
tok (paper's rule)	0.25	1.10	1.02	10 / 90	1.05	78%
tok_It (ours)	0.15	1.13	1.01	6 / 90	1.08	79%
tok_It (ours)	0.2	1.13	1.04	6 / 90	1.06	78%
tok_It (ours)	0.25	1.13	1.07	11 / 90	1.01	66%

Shaded row: best tok_It setting (fewest verifier passes). tok rows: the paper's rule at the tightest and loosest of the three settings.



Left: mean answer length vs mean accepted length per pass (x-axis), all settings of both rules; lossless is the x. Right: accuracy vs the same x-axis.

Long reasoning; 6 of 90 lossless runs loop to 32k tokens and dominate the means. tok_It $\alpha=0.15$: same length, same loop rate, +1 accuracy point, 8% fewer passes. tok_It $\alpha=0.25$ loses accuracy (66%).

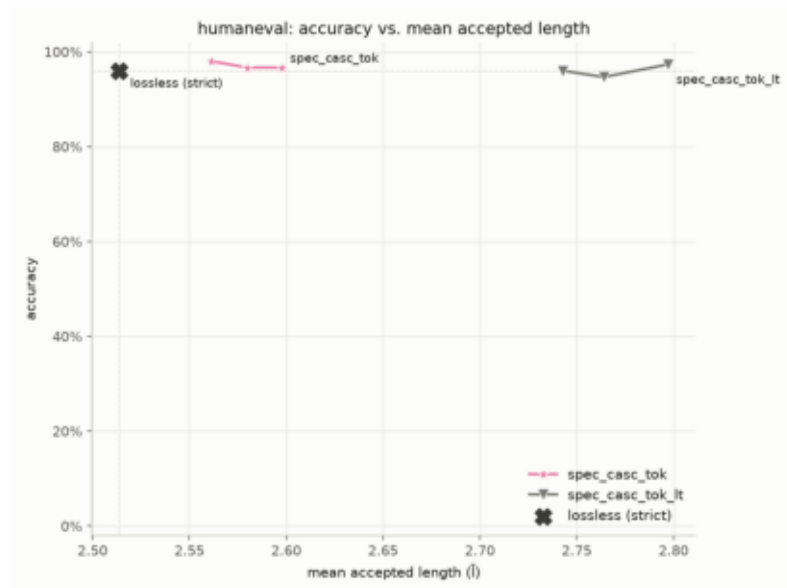
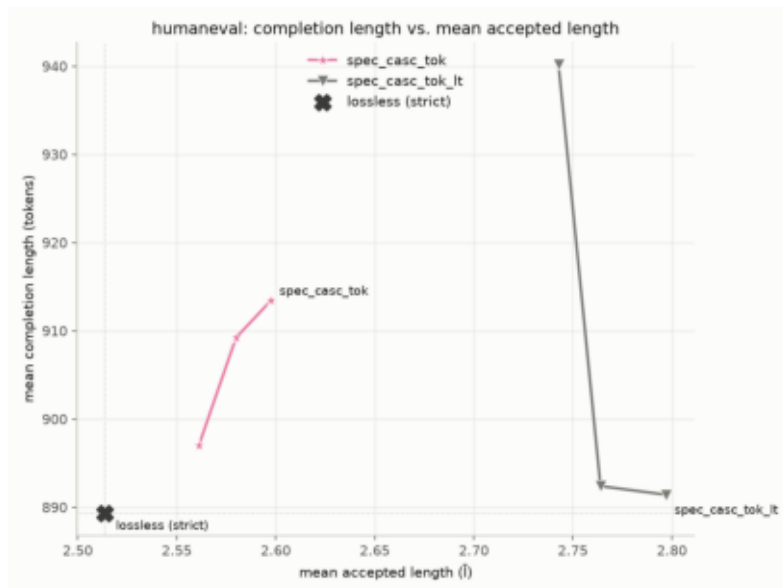
HumanEval — code generation

150 problems × 3 seeds = 450 runs per point · budget 9,000 tokens · relative to lossless, paired per problem and seed

lossless baseline: 2.51 accepted per pass, 889 tokens per answer, 1 of 450 runs hit the budget, accuracy 96%

rule	α	accepted per pass	answer length	budget hits	verifier passes (speedup)	accuracy
lossless	—	1.00	1.00	1 / 450	1.00	96%
tok (paper's rule)	0.15	1.03	1.02	1 / 450	1.00	96%
tok (paper's rule)	0.25	1.03	1.03	0 / 450	1.01	96%
tok_lt (ours)	0.15	1.10	1.00	0 / 450	1.08	95%
tok_lt (ours)	0.2	1.09	1.06	2 / 450	1.01	96%
tok_lt (ours)	0.25	1.11	1.00	0 / 450	1.09	97%

Shaded row: best tok_lt setting (fewest verifier passes). tok rows: the paper's rule at the tightest and loosest of the three settings.



Left: mean answer length vs mean accepted length per pass (x-axis), all settings of both rules; lossless is the x. Right: accuracy vs the same x-axis.

tok_lt gains 8-9% at α 0.15/0.25 with lossless length and accuracy; tok gains nothing end to end (≈ 1.00 - $1.01\times$).

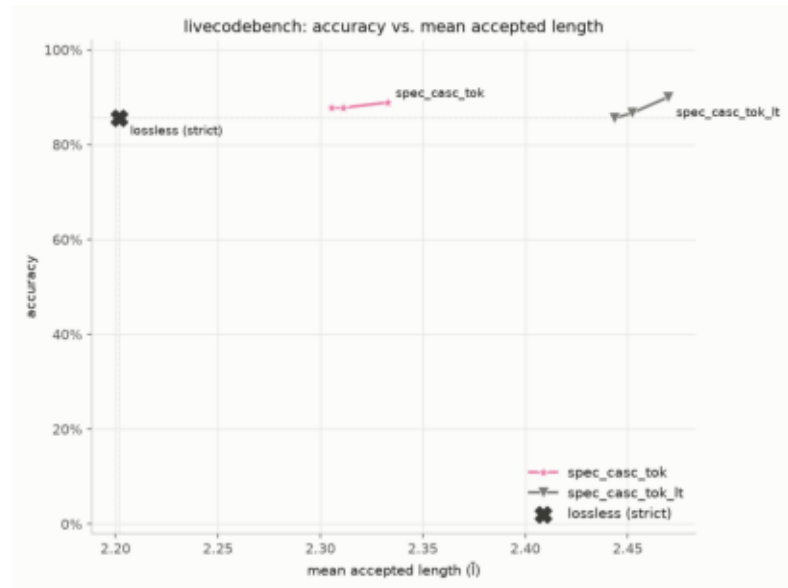
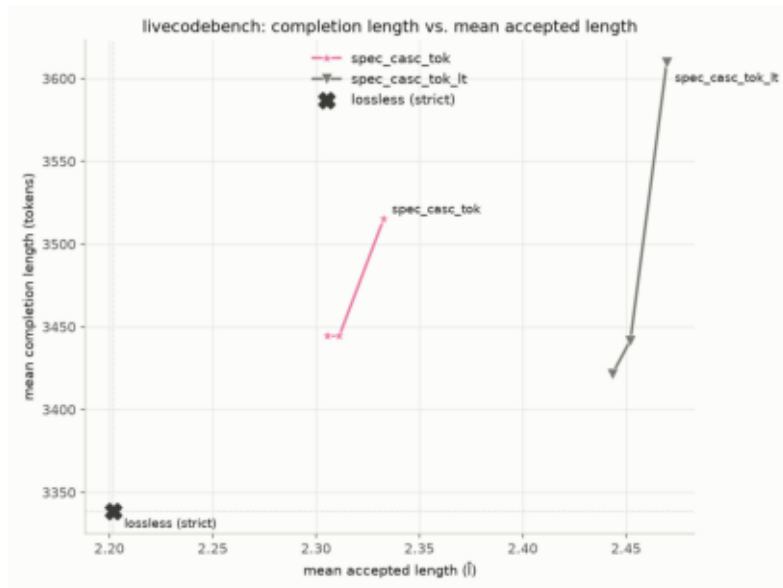
LiveCodeBench — competitive programming

90 problems × 3 seeds = 270 runs per point · budget 12,000 tokens · relative to lossless, paired per problem and seed

lossless baseline: 2.20 accepted per pass, 3,339 tokens per answer, 9 of 270 runs hit the budget, accuracy 89%

rule	α	accepted per pass	answer length	budget hits	verifier passes (speedup)	accuracy
lossless	—	1.00	1.00	9 / 270	1.00	89%
tok (paper's rule)	0.15	1.05	1.03	12 / 270	1.01	93%
tok (paper's rule)	0.25	1.06	1.05	12 / 270	1.00	89%
tok_lt (ours)	0.15	1.11	1.02	10 / 270	1.06	89%
tok_lt (ours)	0.2	1.11	1.03	10 / 270	1.06	90%
tok_lt (ours)	0.25	1.12	1.08	14 / 270	1.01	89%

Shaded row: best tok_lt setting (fewest verifier passes). tok rows: the paper's rule at the tightest and loosest of the three settings.



Left: mean answer length vs mean accepted length per pass (x-axis), all settings of both rules; lossless is the ×. Right: accuracy vs the same x-axis.

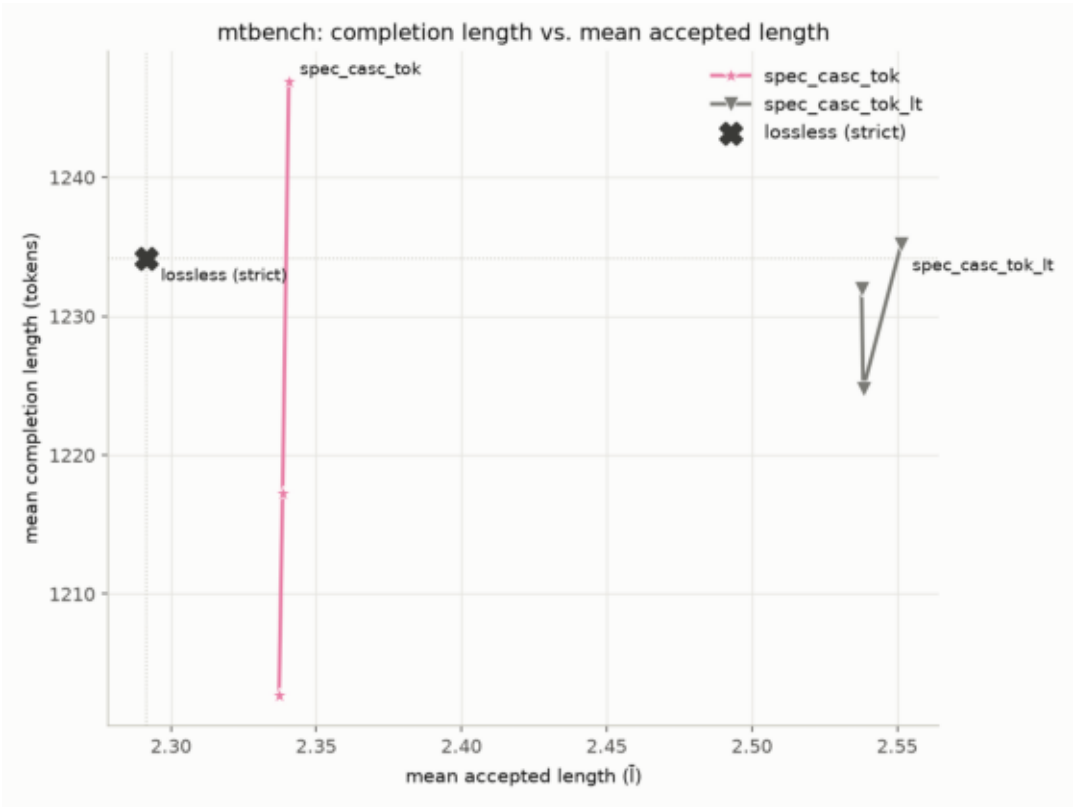
tok_lt 1.06× at α 0.15/0.20; both rules keep accuracy at or above lossless (89%).

MT-Bench — chat (no accuracy grader)

80 problems × 3 seeds = 240 runs per point · budget 4,096 tokens · relative to lossless, paired per problem and seed
lossless baseline: 2.29 accepted per pass, 1,234 tokens per answer, 4 of 240 runs hit the budget, accuracy —

rule	α	accepted per pass	answer length	budget hits	verifier passes (speedup)	accuracy
lossless	—	1.00	1.00	4 / 240	1.00	—
tok (paper's rule)	0.15	1.02	0.99	5 / 240	1.02	—
tok (paper's rule)	0.25	1.02	0.97	5 / 240	1.04	—
tok_lt (ours)	0.15	1.11	1.00	5 / 240	1.08	—
tok_lt (ours)	0.2	1.11	1.00	5 / 240	1.09	—
tok_lt (ours)	0.25	1.11	0.99	2 / 240	1.10	—

Shaded row: best tok_lt setting (fewest verifier passes). tok rows: the paper's rule at the tightest and loosest of the three settings.



Mean answer length vs mean accepted length per pass, all settings of both rules; lossless is the × (no accuracy grader for this benchmark).

No accuracy grader (needs an LLM judge). tok_lt 1.08–1.10× at lossless length; tok ≈1.00–1.04×.

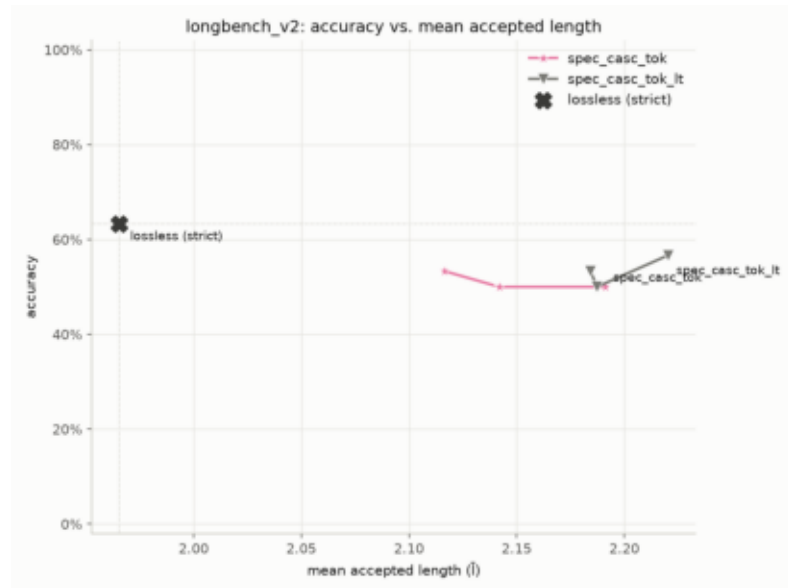
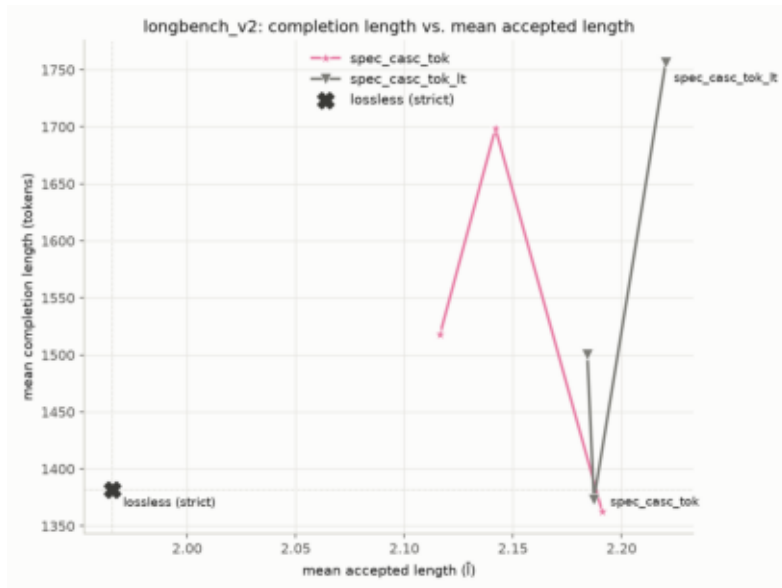
LongBench-v2 — long-context QA

30 problems × 3 seeds = 90 runs per point · budget 8,192 tokens · relative to lossless, paired per problem and seed

lossless baseline: 1.97 accepted per pass, 1,382 tokens per answer, 0 of 90 runs hit the budget, accuracy 60%

rule	α	accepted per pass	answer length	budget hits	verifier passes (speedup)	accuracy
lossless	—	1.00	1.00	0 / 90	1.00	60%
tok (paper's rule)	0.15	1.08	1.10	0 / 90	0.95	56%
tok (paper's rule)	0.25	1.11	0.99	0 / 90	1.09	52%
tok_lt (ours)	0.15	1.11	1.09	1 / 90	1.00	56%
tok_lt (ours)	0.2	1.11	0.99	0 / 90	1.09	53%
tok_lt (ours)	0.25	1.13	1.27	5 / 90	0.88	56%

Shaded row: best tok_lt setting (fewest verifier passes). tok rows: the paper's rule at the tightest and loosest of the three settings.



Left: mean answer length vs mean accepted length per pass (x-axis), all settings of both rules; lossless is the ×. Right: accuracy vs the same x-axis.

The exception. 30 problems only (150 would take ~66 h). No consistent speedup (per-seed 0.63–1.40) and every relaxed setting 3–8 points below lossless's 60% — the one task where the safe rules are not free, in line with the campaign's 150-problem result.

Notes and caveats

Data

- All runs on Nibi (Alliance), one H100 SXM 80 GB per job, vLLM 0.26.0, GPT-OSS-20B with the nebius EAGLE3 drafter, temperature 1.0, top-p 1.0, 6 drafted tokens per pass, seeds 0/1/2.
- One warm vLLM server per (rule, α , seed) — request order on a warm engine is a documented (small) confound; the campaign's own paper tables used a fresh server per measurement. Use the 3-seed means, never a single seed.
- Rows are paired: each run is compared with the lossless run of the same problem and the same seed.
- Accuracy graders: gsm8k / AIME24 / LongBench-v2 exact-answer match, HumanEval / LiveCodeBench code execution; MT-Bench has none. A report bug that applied one seed's verdict to all three seeds was fixed on 2026-09-16 and every multi-seed report regenerated; all accuracies here are the corrected per-seed ones.

Reading the numbers

- A rule only helps if the speedup column is above 1.00 while the answer-length column stays near 1.00 and accuracy is unchanged. The source papers report only the acceptance column, which is why their gains do not survive this test.
- AIME24 and gsm8k means are dominated by the few runs that hit the token budget (each is 2-32k tokens); the budget-hit column is the loop rate. Differences of ± 3 loops per 90-450 runs are noise.
- LongBench-v2 uses 30 problems (its answers take ~ 73 s each); accuracy resolution there is coarse (± 5 points).

What was not done

- Wall-clock timing under a clean protocol (verifier passes are the cost proxy); Qwen3-8B (the new rules exist only for vLLM's V1 runner); fresh-server-per-measurement re-runs of these sweeps.

Files: cascade/RESULTS.md (report), cascade/results/per_benchmark_summary.md (every rule $\times$ benchmark), cascade/JOURNAL.md (dated log incl. failures and fixes), campaign/{tables,results,graphs}/<benchmark>_proper.* and aime24_fine.* (raw).